

SKEMA JAWAPAN DAN PEMARKAHAN UJIAN PRA PSPM SET 1, PHYSICS SEMESTER 1, SESI 2023/2024

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KOLEJ MATRIKULASI PERAK
KEMENTERIAN PENDIDIKAN MALAYSIA

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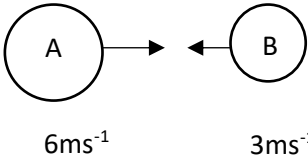
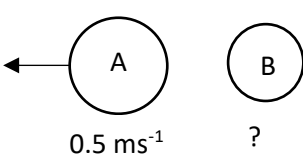
	$\overline{OA - OB} = \sqrt{9.8^2 + -2.7^2}$ $= 10.17 \text{ m}$ $\tan \theta = \frac{R_y}{R_x}$ $\tan \theta = \frac{-2.70}{9.80}$ $\theta = -15.4^\circ \text{ below positive-x axis}$	GJU1
	TOTAL	8M

NO.	ANSWER SCHEME	MARK(S)
2 (a) (i)	For 0-4 s : move 40m to the right For 4-12 s : Stay or not moving For 12-16 s : Move 40 m to the left (to the original position)	J1 J1 J1
2 (a) (ii)	Velocity ms^{-1} Time(s) Distance travelled = $40 + 0 + 40 = 80 \text{ m}$	D2 JU1
2 (b) (i)	$u = 30\text{ms}^{-1}$ $v^2 = u^2 - 2gs$ $0 = 30^2 - 2(9.81)s$ $s = 45.87\text{m}$	K1 for $v = 0$ G1 JU1

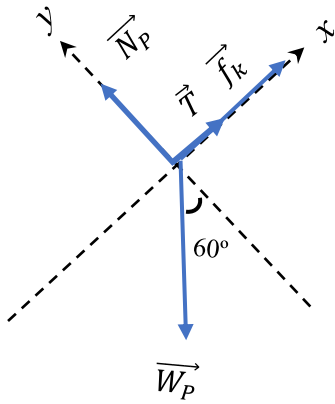
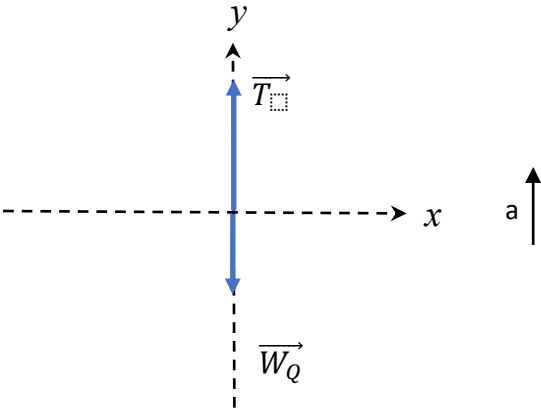
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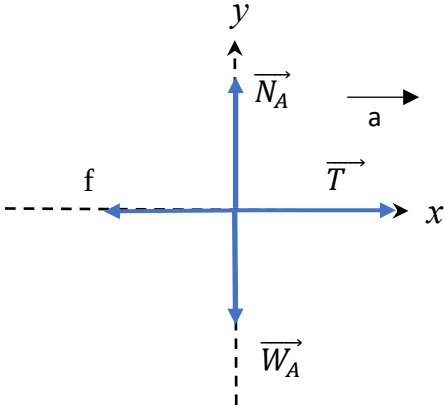
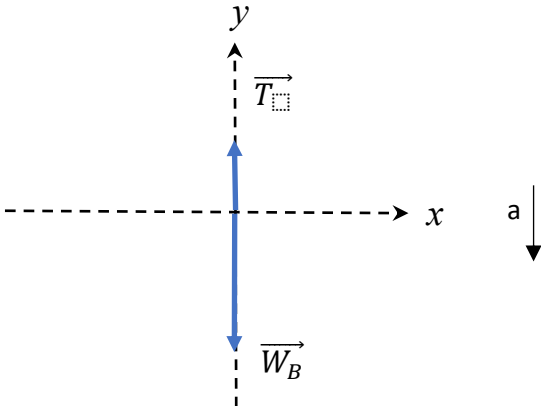
2 (b) (ii)	$v = u - gt$ $0 = 30 - (9.81)t$ $t = 3.06s.$	K1 for $v = 0$ G1 JU1
TOTAL		12M

NO.	ANSWER SCHEME	MARK(S)
3 (a) (i)	$P_A = m_A U_A$ $= (1) (6)$ $= 6kgms^{-1}$	G1 JU1
3 (a) (ii)	Before :  After:  $\sum p_i = \sum p_f$ $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ $(1)(6) + (0.6)(-3) = (1)(-0.5) + (0.6)v_2$ $V_2 = +7.83 \text{ ms}^{-1}$	G1 JU1 J1 – mention the motion to the right.
3 (b) (i)	$J = m(v - u)$ $= 0.5(-14 - 18)$ $= -16Ns$	G1 JU1
3 (b) (ii)	$F = \frac{J}{t}$ $= -\frac{16}{0.06}$ $= 266.67N$	G1 JU1
TOTAL		9M

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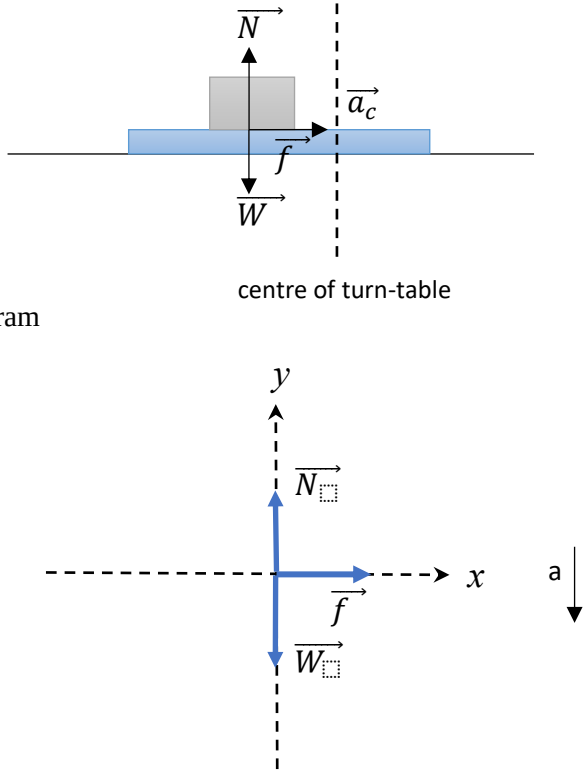
NO.	ANSWER SCHEME	MARK(S)
4 (a)	Free body diagram Box P  $\sum F_x = ma$ $W_p \sin 60^\circ - T - f = W_p(a)$ $(5)(9.81)\sin 60^\circ - T - 5 = 5(a) \quad T + 5a + 37.48 \dots\dots\dots(1)$ Free body diagram box Q  Box Q: $\sum F_y = ma$ $T - W_Q = m_Q a$ $T - (2)(9.81) = (2)(a)$ $T - 2a = 19.62 \dots\dots\dots(2)$ Simultaneous equations of (1) and (2), $T = 24.72 \text{ N}$ $a = 2.55 \text{ m s}^{-2}$	D3 (all forces with label) G1 G1 JU1

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4(b)	Free body diagram Block A  $\sum F_x = ma$ $T - f = m_A a$ $T - \mu_k N_A = m_A a$ $T - (0.15)(78.48) = 8a$ $T - 8a = 11.77 \dots \dots \dots (1)$ Free body diagram box B  $\sum F_y = ma$ $W_B - T = m_B a$ $4(9.81) - T = (4)(a)$ $T + 4a = 39.24 \dots \dots \dots (2)$ Simultaneous equations of (1) and (2), $T = 30.08 \text{ N}$ $a = 2.29 \text{ m s}^{-2}$	D2 (all forces with label) G1 G1 G1 JU1
	TOTAL	12M

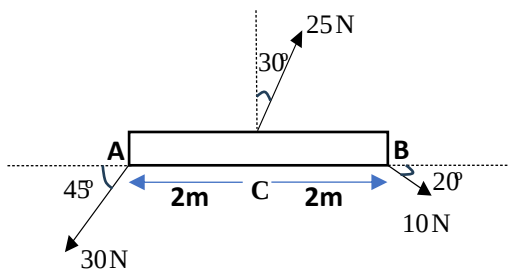
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NO.	ANSWER SCHEME	MARK(S)
5 (a) (i)	$v = r\omega$ $0.75 = (1.2)\omega$ $\omega = 0.63 \text{ rads}^{-1}$	G1 JU1
5 (a) (ii)	$\omega = \frac{\theta}{t}$ $\theta = \omega t$ $= (0.63)(120)$ $= 75.6 \text{ rads}^{-1}$ $\frac{75.6}{2\pi} = 12.03 \text{ rev}$	G1 GJU1
5 (b) (i)	Free body diagram  centre of turn-table $\sum F_x = F_c$ $f = mr\omega^2$ $0.55 = (0.1)(0.5)\omega^2$ $\omega = 3.32 \text{ rads}^{-1}$	D3 (all forces with label) G1 JU1
	TOTAL	9M

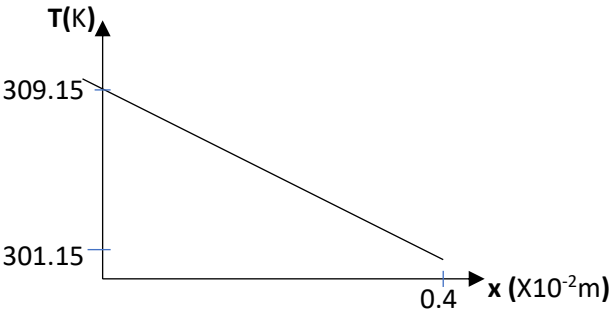
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NO.	ANSWER SCHEME	MARK(S)
6 (a) (i)	$r = 25\text{cm} = 0.25\text{m}$ $\text{when } t = 3\text{s}, \theta = 37 \text{ rev}$ $\omega = 98 \text{ rads}^{-1}$ $\omega_o = \frac{\theta}{t} = \frac{37 \text{ rev}}{3} = \frac{37 \times 2\pi}{3}$ $= 77.49 \text{ rads}^{-1}$	G1 JU1
6 (a) (ii)	$\omega = \omega_o + \alpha t$ $98 = 77.49 + \alpha(3)$ $\alpha = 6.84 \text{ rads}^{-2}$	G1 JU1
6 (a)(iii)	$a = r\alpha$ $= (0.25)(6.84)$ $= 1.71 \text{ ms}^{-2}$	G1 JU1
6(a)(iv)	$\theta = \frac{1}{2}(\omega_o + \omega)t$ $= \frac{1}{2}(77.49 + 98)(3)$ $= 263.24 \text{ rad}$	G1 JU1
6(b)	 Taking moment about the centre of the beam, C ; $\tau_A = r_A F l_A$ $= 2(30 \sin 45^\circ)$ $= +42.43 \text{ Nm}$ $\tau_B = r_B F l_B$ $= 2(10 \sin 20^\circ)$ $= -6.84 \text{ Nm}$ $\tau_{net} = \tau_A + \tau_B$ $= (+42.43) + (-6.84)$ $= +35.59 \text{ Nm}$ (anticlockwise)	G1 G1 G1 JU1
	TOTAL	12M

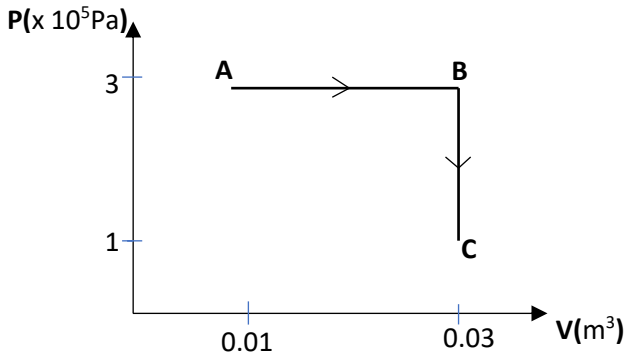
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7 (a) (i)	$A = 1 \times 2 = 2m^2$ $x = 0.40cm = 0.4 \times 10^{-2}m$ $T_f = 28^\circ C = 28 + 273.15 = 301.15K$ $T_i = 36^\circ C = 36 + 273.15 = 309.15K$ $dt = 12h = 12 \times 60 \times 60 = 43200s$ $\frac{dQ}{dt} = -kA \frac{dT}{x}$ $\frac{dQ}{43200} = -(0.9)(2) \left[\frac{301.15 - 309.15}{0.4 \times 10^{-2}} \right]$ $dQ = 1.56 \times 10^8 J$	G1 G1 G1 JU1
7 (a) (ii)		D2 Shapes- D1 Axis & label -D1
7 (b)(i)	$V_{rms} = \sqrt{\langle v^2 \rangle}$ $3.54 \times 10^4 = \sqrt{\frac{(3.5 \times 10^4)^2 + (3.0 \times 10^4)^2 + (3.6 \times 10^4)^2 + v_4^2}{4}}$ $4(3.54 \times 10^4)^2 = (3.5 \times 10^4)^2 + (3.0 \times 10^4)^2 + v_4^2$ $v_4^2 = 1591640000$ $v_4 = 3.99 \times 10^4 ms^{-1}$	G1 JU1
7(b)(ii)	$v_{rms} = \sqrt{\frac{3RT}{M}}$ $4.72 \times 10^2 = \sqrt{\frac{3(8.314)T}{44 \times 10^{-3}}}$ $T = 393.01K$	G1 JU1

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7(b)(iii)	$v_{rms} = \sqrt{\frac{3kT}{m}}$ $v_{rms} \propto \sqrt{T}$ $\frac{v_{rms1}}{\sqrt{T_1}} = \frac{v_{rms2}}{\sqrt{T_2}}$ $v_{rms2} = \left[\frac{\sqrt{T_2}}{\sqrt{T_1}} \right] v_{rms1}$ when $T_2 = 1.4T_1$, $v_{rms2} = \left[\frac{\sqrt{1.4T_1}}{\sqrt{T_1}} \right] v_{rms1}$ $= 1.18v_{rms1}$ $\therefore$ percentage of increase in $v_{rms} = 18\%$	G1 J1
7(c)(i)	 Process AB : Isobaric BC : Isochoric	J1 J1
7(c)(ii)	$T_1 = 40^\circ\text{C} = 313.15\text{K}$ $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $T_2 \left[\frac{V_2}{V_1} \right] T_1$ $\left[\frac{0.03}{0.01} \right] (313.15)$ 939.45K	GJU1
7(c)(iii)	$W_{AC} = W_{AB} + W_{BC}$ $6000 + 0$ 6000J	GJU1

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7(c)(iv)	$\Delta U_{AC} \propto \Delta T_{AC}$ $\text{since } T_A = T_c$ $\Delta T_{AC} = 0$ $\therefore \Delta U_{AC} = 0J$	GJU1
7(c)(v)	$\Delta U_{AC} = Q_{AC} - W_{AC}$ $0 = Q_{AC} - 6000J$ $Q_{AC} = -6000J$	GJU1
	TOTAL	18M